

SIMATS ENGINEERING

SAVEETHA SCHOOL OF ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL – 2

SCENARIO BASED ASSESMENT

Course Code DSA0109

Course Title: Object Oriented Programming
using C++

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**CO Statement Design C++ programs by applying
functions and object-oriented**

**concepts such as classes, objects, access specifiers,
member functions, static members, and arrays of
objects to solve real-**

1)A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a Consultation Charge System. Design a C++ program for hospital management.

Sol)

Input:

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 // Function Prototypes
6 double calculateCharge(double consultationFee, double serviceCharge = 100);
7 double calculateCharge(double consultationFee, double specialistFee, double serviceCharge);
8 void modifyBill(double &bill);
9
10 int main()
11 {
12     int choice;
13     double consultationFee, specialistFee, totalBill;
14     string patientType;
15
16     cout << "==== Hospital Consultation Charge System =====< endl;
17     cout << "1. General Consultation" << endl;
18     cout << "2. Specialist Consultation" << endl;
19     cout << "Enter your choice: ";
20     cin >> choice;
21
22     if (choice == 1)
23     {
24         patientType = "General Patient";
25
26         cout << "Enter General Consultation Fee: ";
27         cin >> consultationFee;
28
29         // Calls overloaded function with default service charge
30         totalBill = calculateCharge(consultationFee);
31     }
32     else if (choice == 2)
33     {
```

Output:

```
==== Hospital Consultation Charge System =====
1. General Consultation
2. Specialist Consultation
Enter your choice: 2
Enter Consultation Fee: 500
Enter Specialist Fee: 500
Enter Additional Charges (0 if none): 100

Patient Type : Specialist Patient
Total Consultation Charge : Rs. 1200
```

2. Design a C++ program Fitness Centre Membership System. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

Input:

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Member
6 {
7 private:
8     int memberID;
9     string name;
10    string planType;
11    double fee;
12
13 public:
14     static int activeMembers;
15
16     // Function to input member details
17     void assignPlan()
18     {
19         cout << "\nEnter Member ID: ";
20         cin >> memberID;
21
22         cout << "Enter Member Name: ";
23         cin.ignore();
24         getline(cin, name);
25
26         cout << "Enter Plan (Basic/Premium/VIP): ";
27         cin >> planType;
28
29         // Validation
30         if (planType == "Basic" || planType == "basic")
31             fee = 1000;
32         else if (planType == "Premium" || planType == "premium")
```

Output:

```
Enter Number of Members: 2
----- Member 1 -----
Enter Member ID: 5246
Enter Member Name: ganesh
Enter Plan (Basic/Premium/VIP): vip
----- Member 2 -----
Enter Member ID: 5246
Enter Member Name: gani
Enter Plan (Basic/Premium/VIP): vip
===== Membership Details =====
Member ID      : 5246
Name           : ganesh
Plan Type      : vip
Fee            : Rs. 3000
Membership Fee: Rs. 3000
Member ID      : 5246
Name           : gani
Plan Type      : vip
Fee            : Rs. 3000
Membership Fee: Rs. 3000
Total Active Members: 2
```

3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System.

Input:

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4
5 // Function Prototype (EMI Calculation)
6 double calculateEMI(double principal, double rate = 7.5, int time = 1);
7
8 // Inline function for Simple Interest
9 inline double calculateSimpleInterest(double principal, double rate, int time) {
10     return (principal * rate * time) / 100;
11 }
12
13 // Function using Call by Reference to apply subsidy
14 void applySubsidy(double &amount, double subsidy) {
15     amount -= subsidy;
16     if (amount < 0) amount = 0;
17 }
18
19 // EMI Calculation Function Definition
20 double calculateEMI(double principal, double rate, int time) {
21     double monthlyRate = rate / (12 * 100);
22     int months = time * 12;
23
24     double emi = (principal * monthlyRate * pow(1 + monthlyRate, months)) /
```

Output:

```
Enter Loan Amount: 5000
Enter Interest Rate (Press 0 for default 7.5%): 2
Enter Loan Tenure (in years): 1
Enter Subsidy Amount: 1000

===== Loan Repayment Details =====
Loan Amount after Subsidy: 4000
Interest Rate: 2%
Tenure: 1 years
Simple Interest: 80
Monthly EMI: 336.955
Total Payment (EMI * Months): 4043.47
```

4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement

a C++ program using functions and object-oriented programming concepts to develop a Parcel Charge Management System.

Input:

```
1  #include <iostream>
2  using namespace std;
3
4  class Parcel {
5  private:
6      string parcelType;
7      double weight;
8      double distance;
9      double packagingCharge;
10
11      static int totalParcels; // static data member
12
13      // Inline function for base charge calculation
14      inline double baseCharge() {
15          return weight * distance * 0.05; // base rate
16      }
17
18  public:
19      // Function to read parcel details (with default argument)
20      void inputDetails(double packCharge = 50) {
21          cout << "Enter Parcel Type (Normal/Express): ";
22          cin >> parcelType;
23
24          cout << "Enter Weight (kg): ";
25          cin >> weight;
26
27          cout << "Enter Distance (km): ";
```

Output:

```
--- Parcel Details ---
Type: normal
Weight: 5 kg
Distance: 100 km
Packaging Charge: 50
Total Delivery Charge: 75

--- Parcel Details ---
Type: express
Weight: 6 kg
Distance: 100 km
Packaging Charge: 50
Total Delivery Charge: 180

Total Parcels Processed: 2
```

5. A utility company wants to automate the calculation of water bills for domestic and commercial consumers. Design and implement a C++ program using functions.

Input:

```

1  #include <iostream>
2  using namespace std;
3
4  class Consumer {
5  private:
6      int consumerID;
7      string consumerType;
8      double consumption;    // in liters
9      double billAmount;
10
11     static int totalConsumers; // static data member
12
13 public:
14     // Function Prototypes
15     void inputDetails();
16     void calculateBill(double taxRate = 0.05); // default argument (service tax)
17     double calculateBill(double rate, double taxRate); // overloaded function
18     void applySubsidy(double &bill, double subsidy); // call by reference
19     void applySubsidyWrapper(double subsidy); // safe wrapper
20     void display();
21     static void showTotalConsumers();
22 };
23
24 // Initialize static member
25 int Consumer::totalConsumers = 0;
26
27 // Input function
28 void Consumer::inputDetails() {
29     cout << "Enter Consumer ID: ";
30     cin >> consumerID;
31
32     cout << "Enter Consumer Type (Domestic/Commercial): ";
33     cin >> consumerType;

```

Output:

```

Enter number of consumers: 1

Enter details for Consumer 1:
Enter Consumer ID: 5246
Enter Consumer Type (Domestic/Commercial): domestic
Enter Water Consumption (liters): 50
Enter subsidy amount: 40

--- Consumer Bill Details ---
Consumer ID: 5246
Consumer Type: domestic
Water Consumption: 50 liters
Final Bill Amount (after subsidy): 0

Total Consumers Processed: 1

```